

LINUX CRASH COURSE

Getting Help and Date/Time

- Getting a short Linux command description and usage information
 - Type: `man <command>`
 - Then use Page-Up/Page-Down and the Up/Down arrows to navigate and “q” to quit
 - E.g.:

```
[centos@ip-172-31-2-119 ~]$ man ls
[centos@ip-172-31-2-119 ~]$
```
- Displaying the current date and time
 - Type: `date`
 - Shows the date and time in UTC (also GMT)

Displaying Text Files 1/2

- Displaying the beginning (first few lines) of a text file:
 - Type: `head <filename>`
- Displaying the end (last few lines) of a text file:
 - Type: `tail <filename>`
 - Useful options:

Option	Effect
<code>-n <number></code>	Displays the last <number> of lines
<code>-f</code>	Follows the file (displays more output as the file grows)

Displaying Text Files 2/2

- Displaying the whole text file:
 - Type: `cat <filename>`
- Displaying a text file page by page:
 - Type: `less <filename>`

- E.g.:

```
[centos@ip-172-31-2-119 ~]$ tail -n 5 /etc/passwd
cockpit-wsinstance:x:993:990:User for cockpit-ws
instances:/nonexisting:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/var/empty/sshd:/sbin/nologin
chrony:x:992:989::/var/lib/chrony:/sbin/nologin
rngd:x:991:988:Random Number Generator
Daemon:/var/lib/rngd:/sbin/nologin
centos:x:1000:1000:Cloud User:/home/centos:/bin/bash
[centos@ip-172-31-2-119 ~]$
```

File System's Directory Hierarchy 1/2

- **File system:** tree-like structure where each directory may contain zero or more files and/or sub-directories
- **Root of the file system:** / (slash)
- **Absolute path:** starts at the root directory / and describes a sequence of sub-directories to reach a file (e.g. absolute path to the current log file: /var/log/messages)
- **Relative path:** describes a sequence of sub-directories to reach a file from the current directory (e.g. if the current directory is /var, the relative path is log/messages)

File System's Directory Hierarchy 2/2

- **Shortcut notation:**
 - Current directory: `.` (dot), one directory level up: `..` (two dots)
 - Current user's home directory: `~` (tilde)
- **Most important directories:**

Directory	Description
<code>/root</code>	The home directory of the administrative superuser (the root)
<code>/usr</code>	Installed SW, shared libraries and other read-only program data
<code>/etc</code>	Configuration files
<code>/var</code>	Variable-size data (e.g. databases, log files, website content)
<code>/run</code>	Runtime process data (deleted on each reboot)
<code>/home</code>	Regular users' home directories (e.g. <code>/home/centos</code>)
<code>/tmp</code>	Temporary files (regularly cleaned up)

File Management Commands 1/3

- Displaying the current working directory:
 - Type: `pwd`
- Changing the current working directory:
 - Type: `cd <new directory>` (can be absolute or relative path)
 - E.g.: `cd /var/log` or `cd ~` or `cd ..`
- Displaying directories and the content of directories:
 - Type: `ls [<options>] [<directory name>]` ← Optional values
 - Useful options:

Option	Effect
-l	Long listing (file/directory attributes are also shown)
-d	Shows the directory itself instead of its content
-h	Shows file sizes in human readable form (e.g. 35M)

File Management Commands 2/3

- Copying files:
 - Type: `cp <source path> <destination path>`
 - E.g. `cp /var/log/messages ~/backup/`
 - You can use wildcards: e.g. `cp /tmp/*.txt /var/tmp/`
- Moving (relocating) files:
 - Type: `mv <source path> <destination path>`
 - E.g. `mv /tmp/myfile.txt ~/backup/`
 - You can also rename files with this command (e.g. `mv /tmp/myfile.txt /tmp/newfile.txt`)
- Deleting files:
 - Type: `rm <file path>` (e.g. `rm myfile.txt`)

File Management Commands 3/3

- Creating a directory:
 - Type: `mkdir <directory name/path>`
 - E.g.: `mkdir ~/backup/mail` or `mkdir Documents`
 - The parent directory of the new directory must exist beforehand
- Deleting a directory:
 - Type: `rmdir <directory name/path>`
 - E.g.: `rmdir ~/backup/mail` or `rmdir Documents`
 - The directory must be empty before deletion

Simple Text Editor: nano

- **Package name:** nano
 - Install nano: `sudo yum install nano`
 - Start nano: `nano [<file name>]`
- **Common commands/functions:**
 - Copy text: highlight with cursor
 - Paste text: right-click with mouse
 - Delete current line: `CTRL + k`
 - Save file: `CTRL + o`
 - Exit: `CTRL + x`

Users and Groups 1/2

- **User:** an entity that has controlled access to system resources
 - Can be a human or a daemon/system process
 - User ID (UID): a number that identifies the user
 - Local user info is stored in the file `/etc/passwd`
- **Group:** a collection of users who have the same access privileges to some of the system resources
 - Simplifies user management
 - Group ID (GID): a number that identifies a user group
 - Group information is stored in the file `/etc/group`
 - Primary group: has the same name as the user and the user is the only member of this group

Users and Groups 2/2

- Administrative super user:
 - Has all power over the Linux system
 - User name: `root`, home directory: `/root`
- To switch to another user account: `su [<user name>]`
- To temporarily elevate privileges to the root user level:
 - Type: `sudo <command>`
 - E.g.: `sudo yum install nano`
 - The command will be executed with root privileges, but the user does not switch to the root account
 - Preferred way to execute privileged admin tasks
 - In AWS, the user `centos` has `sudo` privileges

File System Permissions 1/4

- Permissions for files and directories:

Permission	Symbol	File	Directory
Read	r	The file can be read/viewed.	The file names in the directory can be listed with the ls command.
Write	w	The file can be modified/edited.	New files can be created in the directory, and existing files can be renamed or deleted (must also have Execute permission).
Execute	x	The file can be executed as a command.	The directory can be entered/traversed with the cd command, and the file attributes (size, time stamp, permissions) can be accessed.

File System Permissions 2/4

- To view file permissions: `ls -l [<directory name>]`
- E.g.:

```
[centos@ip-172-31-2-119 ~]$ ls -l /home/centos
total 4
-rw-rw-r--. 1 centos centos 123 Aug  3 18:23 test.txt
```

File type
-: regular file
d: directory

Permissions for the file's owner (read, write, no execute)

Permissions for the file's owning group (read, write, no execute)

Permissions for all other users (read, no write, no execute)

File's owner

File's owning group

File size in bytes

Last modified date/time

File name

File System Permissions 3/4

- To change file/directory permissions:
 - Type: `chmod <permissions> <file/dir name>`
 - Notation:
 - Entities: owning user: `u`, owning group: `g`, other users: `o`
 - Adding permission: `+`, Removing permission: `-`
 - Examples:
 - Adding write permission for the owning user: `u+w`
 - Adding read, write and execute permission to the owning group: `g+rwX`
 - Removing read permission from other users: `o-r`

File System Permissions 4/4

- For example:

```
[centos@ip-172-31-2-119 ~]$ ls -l /home/centos
total 4
-rw-rw-r--. 1 centos centos 123 Aug  3 18:23 test.txt
[centos@ip-172-31-2-119 ~]$ chmod g-w test.txt
[centos@ip-172-31-2-119 ~]$ ls -l /home/centos
total 4
-rw-r--r--. 1 centos centos 123 Aug  3 18:23 test.txt
[centos@ip-172-31-2-119 ~]$
```


Process Management 1/4

- **Process:** running instance of and executable program or command
- **Process ID (PID):** a unique number that identifies the process
- **Process priority:** governs how much CPU time the process gets
 - Values: 0 ... 39, lower value means higher priority
 - Controlled by the OS process scheduler (dynamically changes)
- **Nice value:** administratively set quantity describing the importance of the process
 - Values: -20 ... +19, lower value means higher importance
 - Default value for user processes is 0
 - Set by the system administrator (static quantity)

Process Management 2/4

- Monitoring processes, CPU and memory usage:
 - Type: **top** (press 'h' for help with display options)
 - 1st line: uptime, number of logged-in users, and load averages for the last 1, 5 and 15 minutes
 - 2nd line: overall process status (running, sleeping, etc.)
 - 3rd line: CPU utilization: user processes (us), system processes (sy)
 - 4th and 5th lines: memory usage (physical and virtual/swap memory)
 - Task area: list of processes with their PIDs, owning user, CPU and memory utilization and the associated command

Process Management 3/4

- Monitoring processes, CPU and memory usage (cont.):
 - Example: Monitoring CPU and memory usage using **top**

```
centos@ip-172-31-60-221:~  
top - 20:44:18 up 6 min, 1 user, load average: 0.01, 0.04, 0.03  
Tasks: 82 total, 1 running, 81 sleeping, 0 stopped, 0 zombie  
%Cpu(s): 0.0 us, 0.0 sy, 0.0 ni, 100.0 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st  
GiB Mem : 1.0 total, 0.8 free, 0.1 used, 0.1 buff/cache  
GiB Swap: 0.0 total, 0.0 free, 0.0 used. 0.7 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
17	root	20	0	0.0m	0.0m	0.0m	S	0.3	0.0	0:00.13	kworker/0:1
1	root	20	0	124.9m	6.4m	4.0m	S	0.0	0.6	0:03.52	systemd
2	root	20	0	0.0m	0.0m	0.0m	S	0.0	0.0	0:00.00	kthreadd
4	root	0	-20	0.0m	0.0m	0.0m	S	0.0	0.0	0:00.00	kworker/0:0H
5	root	20	0	0.0m	0.0m	0.0m	S	0.0	0.0	0:00.00	kworker/u30+
6	root	20	0	0.0m	0.0m	0.0m	S	0.0	0.0	0:00.02	ksoftirqd/0
7	root	rt	0	0.0m	0.0m	0.0m	S	0.0	0.0	0:00.00	migration/0
8	root	20	0	0.0m	0.0m	0.0m	S	0.0	0.0	0:00.00	rcu_bh
9	root	20	0	0.0m	0.0m	0.0m	S	0.0	0.0	0:00.34	rcu_sched
10	root	0	-20	0.0m	0.0m	0.0m	S	0.0	0.0	0:00.00	lru-add-dra+

Process Management 4/4

- Changing process priority (nice value) for a running process:
 - Type: `renice -n <nice value> <PID>`
 - E.g.: `renice -n -7 12170`
 - Regular users can only increase the nice value of their own processes
 - Only the root user can increase/decrease the nice value of any process (use `sudo`)
- Terminating/killing processes:
 - Type: `kill <PID>` (if it does not work, try `kill -9 <PID>`)
 - E.g. `kill 11667`
 - Only the root user can kill other users' processes (use `sudo`)

Monitoring the File System

- Type: `df -h`
- Shows the available disk/SSD partitions and their associated mount points, partition sizes and usage
- E.g.:

```
[centos@ip-172-31-55-58 ~]$ df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        380M   0    380M   0% /dev
tmpfs           408M   0    408M   0% /dev/shm
tmpfs           408M  11M   398M   3% /run
tmpfs           408M   0    408M   0% /sys/fs/cgroup
/dev/xvda2      10G  1.9G   8.2G  19% /
tmpfs           82M   0     82M   0% /run/user/1000
[centos@ip-172-31-55-58 ~]$
```

File system root

Service and Daemon Management 1/2

- **Service:** a collection of processes that run in the background and implement a given functionality (e.g. HTTP server)
 - Only the root user can start/stop services (use **sudo**)
- **To start a service:**
 - Type: **systemctl start <service name>**
 - E.g. **systemctl start httpd**
- **To stop a service:**
 - Type: **systemctl stop <service name>**
- **To check the status of a service:**
 - Type: **systemctl status <service name>**

Service and Daemon Management 2/2

- For example:

```
[centos@ip-172-31-55-58 ~]$ systemctl status httpd
```

- httpd.service - The Apache HTTP Server

```
Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; vendor preset: disabled)
```

```
Active: active (running) since Wed 2021-08-04 15:06:19 UTC; 2min 29s ago
```

```
Docs: man:httpd.service(8)
```

```
Main PID: 13233 (httpd)
```

```
Status: "Running, listening on: port 80"
```

```
Tasks: 213 (limit: 4855)
```

```
Memory: 24.9M
```

```
. . .
```

Monitoring the System Log

- Important events from the processes and the kernel are recorded in log files in the `/var/log` directory
- Most standard log messages are stored in the file `/var/log/messages`
- To monitor a log file in real time:
 - Type: `tail -f /var/log/messages` (needs root privileges)

```
[centos@ip-172-31-55-58 ~]$ sudo tail -f /var/log/messages
Aug  4 15:05:24 ip-172-31-55-58 systemd[1]: Started /usr/bin/systemctl start
man-db-cache-update.
Aug  4 15:05:24 ip-172-31-55-58 systemd[1]: cgroup compatibility translation
between legacy and unified hierarchy settings activated. See cgroup-compat
debug messages for details.
. . .
```


Software Installation 1/2

- The `yum` package manager automatically resolves package dependencies and installs packages the current package depends on (specific to Fedora/Red Hat/CentOS Linux versions)
- To list the installed and available packages:
 - Type: `yum list`
- To view detailed information about a package:
 - Type: `yum info <package name>`
 - E.g.: `yum info postgresql`

Software Installation 2/2

- To install a package:
 - Type: `yum install <package name>` (needs root privileges)
 - E.g. `yum install postgresql`
 - Packages that this package depends on will also be installed
- To update all installed packages to the latest version
 - Type: `yum update` (needs root privileges)
 - Removes old versions of the updated packages
- To remove a package:
 - Type: `yum remove <package name>` (needs root privileges)
 - Also removes all packages that depend on this package